34 · Male versus female: the comparison table

The words of this track, as it says them. Bold brown is the examiner.

The male versus female table. Six marks.

Tabulate the differences between the male and female reproductive systems, looking at: the function of LH, the function of FSH, the timing of gamete production, the size of the gametes, the timing of meiosis, and the mobility of the gametes.

Six rows, two columns, and one line in each box. Every row has to be a contrast. A row with the same words in both columns scores nothing.

Row one, LH. In females, it triggers ovulation, and forms the corpus luteum. In males, it stimulates the Leydig cells to make testosterone.

Row two, FSH. In females, it stimulates follicle development. In males, it stimulates spermatogenesis, through the Sertoli cells.

Row three, the timing of gamete production. Females start in fetal life, and release eggs from puberty to menopause. Males start at puberty, and keep going for life.

Row four, size. The egg is the largest cell in the body, full of cytoplasm. The sperm is the smallest, with almost none.

Row five, the timing of meiosis. In males it's continuous, and one spermatocyte gives four sperm. In females it stops and starts, arresting in prophase one and again in metaphase two, and one oocyte gives one egg, plus polar bodies.

Row six, mobility. Sperm are motile: they swim with a tail. The egg can't move itself. It's carried along the fallopian tube by cilia, and by peristalsis.

The traps. LH in males goes to the Leydig cells, to make testosterone. Her student-marked feedback slips and says oestrogen. Her Male Repro deck says testosterone, twice. Write testosterone. Second: millions of sperm, but one egg a month. Third: only the sperm moves itself.

Once more. The six rows.

LH: in females it triggers ovulation; in males the Leydig cells make testosterone. FSH: follicle development, or spermatogenesis via the Sertoli cells. Timing: females start in fetal life and release eggs from puberty to menopause; males go from puberty, for life. Size: the egg is the largest cell, the sperm the smallest. Meiosis: in females it stops and starts and gives one egg; in males it's continuous and gives four sperm. Mobility: the egg is carried by cilia; the sperm swims.